

Travel Made Easy for the Visually Impaired with OpenStreetMap

World-Map-Explorer is an open source, screen reader-enabled map built to help the visually impaired and keyboard-first users explore the world.



Picture this: You're planning a trip to Goa, and like millions of others, you pull up a digital map on your phone. You pinch, swipe, scroll, zoom and tap on the online map to explore. In a matter of a few minutes, you've traced the beaches, markets, restaurants, and mentally bookmarked your evening plans.

Now imagine the same scenario, but you can't see the screen. Traditional maps are visual by design, leaving 285 million people worldwide living with visual impairments locked out of one of our most fundamental tools for understanding place and space. This is what the World-Map-Explorer is solving for — an open source product that reimagines digital maps through an inclusive design that uses sound and keyboard navigation for user interaction.

The architecture of inclusion

Behind the product's deceptively

simple interface lies sophisticated technology. The platform leverages Leaflet, Wikidata, Nominatim, Valhalla routing API, Turf.js, Overpass API, etc, drawing from the vast OpenStreetMap database to create something unprecedented: a fully keyboard-navigable map that speaks.

Users navigate with arrow keys, gliding across continents while receiving real-time audio feedback about their location. Press the right arrow to head east from Berlin, and the system announces your new coordinates, elevation changes, and proximity to national borders. Search for the Amazon River, and the platform guides you along its path using nothing but sound and keyboard commands.

Simply by setting a direction and distance, you can simulate movement across a terrain—like pointing a compass and walking, but through pure audio feedback. Behind the scenes, the

system uses mathematical formulas like the Haversine calculation to determine precise geographic positions. To users, it simply feels like exploration.

Now, the community behind World-Map-Explorer is exploring new features such as multilingual voice support, offline access, and compatibility with tactile devices for hands-on exploration. The goal is simple: make geography and the tools to explore it available to everyone.

Open source with a purpose

Unlike most tech products that sit behind paywalls or proprietary walls, World-Map-Explorer is fully open source. That means anyone can view, use, adapt, or contribute to its code. It's a community effort—built not only to solve a technical challenge, but also to make geography accessible to more people.

What adds more meaning to this initiative is that it is inclusively designed—it looks like a regular OpenStreetMap